STAT 140: Design of Experiments

Section 5: Rerandomization

March 2, 2026

By the end of this tutorial, you should understand:

- **Why rerandomization is needed:** even valid randomizations can produce covariate imbalance by chance
- **The six-step procedure** for implementing rerandomization in a real experiment
- **The Mahalanobis distance** M as a scalar measure of multivariate covariate balance, and why it handles correlated covariates better than checking each one individually
- **How to choose the threshold** a using the χ^2_K distribution and the trade-off between balance quality and computational cost
- **Why and when rerandomization improves precision:** why $\text{Var}(\hat{\tau})$ decreases when covariates predict outcomes, and what the variance decomposition tells us
- **Why inference must change** after rerandomization: the randomization test reference set is restricted, and Neymanian intervals involve truncated normals

1 Motivation: Why Rerandomize?

Randomization guarantees covariate balance *on average* across infinitely many experiments. However, in any single experiment, you may get unlucky. Consider the following exchange, recounted by Rubin (2008):

> *Rubin:* What if, in a randomized experiment, the chosen randomized allocation exhibited substantial imbalance on a prognostically important baseline covariate?
>
> *Cochran:* Why didn't you block on that variable?
>
> *Rubin:* Well, there were many baseline covariates, and the correct blocking wasn't obvious.
>
> *Cochran:* This is a question that I once asked Fisher, and his reply was unequivocal:
>
> *Fisher (via Cochran):* Of course, if the experiment had not been started, I would rerandomize.

Fisher's intuition was formalised by Morgan and Rubin (2012): if the randomization you drew is "bad" (meaning it creates noticeable imbalances in important baseline covariates between treatment and control groups) and the experiment has not yet started, you should discard it and draw again. This is **rerandomization**.

Key framing: Rerandomization is not cheating. It is a principled design choice made *before* observing any outcomes, based solely on pre-treatment covariate information. The balance criterion must be specified in advance, and the analysis must account for the restricted randomization distribution.

2 The Rerandomization Procedure

1. **Collect covariate data** on all N units before the experiment begins.
2. **Specify a balance criterion in advance.** This is your formal rule for what counts as an acceptable randomization. You must commit to this *before* randomizing, or you are no longer doing valid inference.
3. **Randomize** the units to treatment and control groups.
4. **Check the balance criterion.** If it is satisfied $\Rightarrow$ go to Step 5. If not $\Rightarrow$ discard this randomization and return to Step 3.
5. **Conduct the experiment** using the accepted randomization.
6. **Analyze the results** using a randomization test, keeping only simulated treatment assignments that *also satisfy the balance criterion from Step 2*. This step is crucial: the inference procedure must mirror the design procedure.

Why must Step 6 match Step 2? The p-value from a randomization test is the proportion of randomizations (under the null) that produce a test statistic at least as extreme as the observed one. However, after rerandomization, the set of "valid" randomizations is no longer all $\binom{N}{N_T}$ possible assignments — it is only those that pass your balance criterion. If you compute p-values using all randomizations, you will get the wrong answer.

3 The Formal Setup

3.1 Notation

We have N units, SUTVA holds, and $W_i \in \{0, 1\}$ is the treatment indicator:

$$W_i = \begin{cases} 1 & \text{if unit } i \text{ is treated,} \\ 0 & \text{if unit } i \text{ is control.} \end{cases}$$

Each unit has potential outcomes $Y_i(0)$ and $Y_i(1)$. The **causal estimand** is:

$$\tau = \overline{Y}(1) - \overline{Y}(0) = \frac{1}{N}\sum_{i=1}^{N} Y_i(1) - \frac{1}{N}\sum_{i=1}^{N} Y_i(0).$$

The observed outcome is $Y_i^{obs} = W_i^{obs}\, Y_i(1) + (1 - W_i^{obs})\, Y_i(0)$, and the **difference-in-means estimator** is:

$$\hat{\tau} = \overline{Y}_T^{obs} - \overline{Y}_C^{obs}, \qquad \overline{Y}_T^{obs} = \frac{1}{N_T} \sum_{i:\, W_i = 1} Y_i^{obs}, \quad \overline{Y}_C^{obs} = \frac{1}{N_C} \sum_{i:\, W_i = 0} Y_i^{obs}.$$

3.2 The Covariate Matrix

Let X be an $N \times K$ matrix of K pre-treatment covariates measured on all units. The columns of X can include:

- original covariates (age, baseline test score, income, ...),
- any functions of original covariates (powers, logs, interactions).

We want the treatment and control groups to look similar on all K columns of X.

3.3 The Acceptance Criterion

The rerandomization criterion is a scalar function of X and W:

$$\varphi(X, W) = \begin{cases} 1 & \text{if } W \text{ is an acceptable randomization,} \\ 0 & \text{otherwise.} \end{cases}$$

$P(\varphi(X, W) = 1) = p_a$ is the **proportion of acceptable randomizations**.

The balance–speed trade-off. A smaller p_a enforces stricter balance (fewer randomizations pass), but also means a longer expected waiting time: on average, you need $1/p_a$ attempts to find one acceptable randomization. If $p_a = 0.001$, you expect $\sim$1,000 draws before one passes.

4 The Mahalanobis Distance

4.1 Why Not Check Each Covariate Individually?

If covariates were uncorrelated and on the same scale, we could just check whether each standardised mean difference $(\bar{X}_{jT} - \bar{X}_{jC})$ is small. However, in practice:

- Covariates are often correlated (age and health, vocabulary and math scores, etc.).
- Checking each covariate individually with a square acceptance region ignores the joint distribution of imbalances.

Figure below illustrates this problem directly. Each dot represents one simulated randomization, plotted at coordinates $(\bar{X}_{1T} - \bar{X}_{1C},\ \bar{X}_{2T} - \bar{X}_{2C})$: its imbalance on covariate 1 (horizontal) and covariate 2 (vertical). A dot at the origin is a perfectly balanced randomization. Both plots show two acceptance regions, both set at $p_a = 0.5$:

- **Red square:** checking each covariate individually with its own threshold. Acceptable means $|\bar{X}_{1T} - \bar{X}_{1C}| \leq c_1$ *and* $|\bar{X}_{2T} - \bar{X}_{2C}| \leq c_2$, forming a rectangle.
- **Blue ellipse:** the Mahalanobis acceptance region $M \leq a$, which accounts for the joint distribution of imbalances.

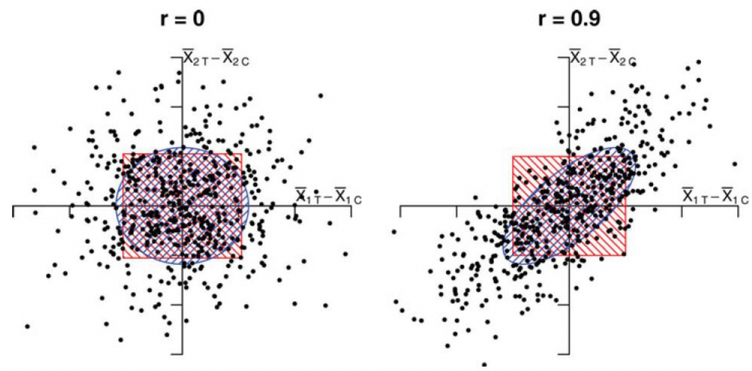

Figure 4. Points displayed are $\overline{X}_{j,T} - \overline{X}_{j,C}$ for two covariates, under orthogonality (left) and correlation (right), under pure randomization of units into two equally sized treatment groups. Acceptance regions are shown for two different rerandomization methods, both with $p_a = 0.5$. The squares are the acceptance regions restricting each covariate individually and the ellipses are the acceptance regions based on M.

Left plot ($r = 0$, covariates uncorrelated) When the two covariates are uncorrelated, the cloud of randomizations is roughly circular. The square and the ellipse are almost the same shape and cover nearly the same set of randomizations. Either approach works fine here.

Right plot ($r = 0.9$, covariates highly correlated) When covariates are strongly correlated, the cloud rotates into a diagonal shape: if X_1 is imbalanced in one direction, X_2 tends to follow. The square stays axis-aligned because it only knows about each covariate separately. The ellipse *rotates* to follow the correlation structure. This creates two problems with the square:

- **The corners.** The square accepts randomizations in its four corners that fall outside the ellipse. These are assignments where the two imbalances point in *opposite* directions (e.g. X_1 negative and large, X_2 positive and large). This is suspicious precisely because highly correlated covariates should be imbalanced together or not at all — opposite-direction imbalances are the unusual, problematic cases the Mahalanobis distance correctly identifies and rejects.
- **The ellipse tails.** The ellipse accepts some randomizations outside the square (along the diagonal) that the square would reject. These are actually fine: both covariates are slightly off in the same correlated direction, which the joint distribution considers acceptable.

What the figure teaches. The square treats covariates as if they live in independent worlds. The ellipse understands that they are correlated and defines balance in the natural geometry of the joint distribution. When $r = 0$, it does not matter which you use. When covariates are correlated – as they almost always are in practice – only the Mahalanobis ellipse gives the correct acceptance region. The square will accept some bad randomizations and reject some good ones.

4.2 Definition

The **Mahalanobis distance** summarises the overall multivariate imbalance as a single scalar:

$$M = M(\overline{X}_T; \overline{X}_C) = (\overline{X}_T - \overline{X}_C)' \left[\text{cov}(\overline{X}_T - \overline{X}_C)\right]^{-1} (\overline{X}_T - \overline{X}_C).$$

Because $\text{cov}(\overline{X}_T - \overline{X}_C) = \left(\frac{1}{N_T} + \frac{1}{N_C}\right) \text{cov}(X)$, this simplifies to:

$$M = \frac{1}{\dfrac{1}{N_T} + \dfrac{1}{N_C}} (\overline{X}_T - \overline{X}_C)' \left[\text{cov}(X)\right]^{-1} (\overline{X}_T - \overline{X}_C).$$

Breaking down the formula. There are three components:

- $(\overline{X}_T - \overline{X}_C)$: the K-vector of covariate mean differences. Perfect balance $\Rightarrow$ this is $\mathbf{0}$ and $M = 0$.
- $[\text{cov}(X)]^{-1}$: rescales and decorrelates the covariates. This is what makes M **affinely invariant** — if you multiply a covariate by a constant (e.g. change units from kg to lbs), M does not change.
- $\dfrac{1}{1/N_T + 1/N_C}$: a harmonic-mean sample-size factor, analogous to the denominator of a two-sample t-statistic.

The rerandomization criterion based on M is:

$$\varphi_M(X, W) = \begin{cases} 1 & \text{if } M \leq a, \\ 0 & \text{otherwise.} \end{cases}$$

Rerandomize whenever $M > a$.

Intuition for affine invariance. An affinely invariant measure treats covariates as defining a shape in K-dimensional space and measures distance in that shape's natural units — it does not matter which axes you choose or how you scale them. Ordinary Euclidean distance is *not* affinely invariant: doubling the scale of one covariate doubles its contribution. The Mahalanobis distance automatically adjusts for scale and correlation, giving a fair composite score.

5 Choosing the Threshold a

5.1 The Null Distribution of M

Under pure randomization and adequate sample sizes, the Mahalanobis distance follows a chi-squared distribution:

$$M \sim \chi^2_K,$$

where K is the number of covariates (columns of X).

Why "under pure randomization"? This result describes the distribution of M *before* any acceptance criterion is applied — i.e., if you drew one randomization completely at random with no filtering. It is the **reference distribution** you use to calibrate the threshold a. Once rerandomization is applied, M follows a truncated version of χ^2_K, restricted to $[0, a]$. The logic is: use $M \sim \chi^2_K$ under pure randomization to find the value of a such that exactly p_a of all randomizations pass, *then* start filtering.

This is exactly the null distribution you use to choose a: set $p_a = P(\chi^2_K \leq a)$ and solve for a:

$$a = F^{-1}_{\chi^2_K}(p_a),$$

where $F^{-1}_{\chi^2_K}$ is the quantile function of the chi-squared distribution with K degrees of freedom.

Example: Shadish data ($K = 10$ covariates, $p_a = 0.001$).

The researchers balanced on 10 pre-treatment covariates (vocabulary pre-test, math pre-test, number correct in math, liking of math, liking of literature, preference for literature, ACT composite score, college GPA, age, sex).

Setting $p_a = 0.001$ means: only accept the 0.1% most balanced randomizations.

$$P\big(\chi^2_{10} \leq a\big) = 0.001 \quad \Longrightarrow \quad a = 1.48.$$

Interpretation: Rerandomize whenever $M > 1.48$. On average, you need $1/0.001 =$ 1,000 draws to find one acceptable randomization.

5.2 Variance Reduction

The key benefit of rerandomization is that it *shrinks the variance* of covariate imbalances, and — when covariates are correlated with the outcome — this translates directly into a more precise estimator of τ.

The ratio of the variance of $\hat{\tau}$ under rerandomization to the variance under pure randomization is:

$$v_a = \frac{2}{K} \times \frac{\Gamma\left(\frac{K}{2} + 1,\ \frac{a}{2}\right)}{\Gamma\left(\frac{K}{2},\ \frac{a}{2}\right)},$$

where $\Gamma(\cdot, \cdot)$ denotes the upper incomplete gamma function. The **percent reduction in variance** is $100(1 - v_a)\%$.

Shadish data (continued): how much do we gain?
With $K = 10$ and $a = 1.48$:

$$v_a = \frac{2}{10} \times \frac{\Gamma(6,\ 0.74)}{\Gamma(5,\ 0.74)} = 0.12.$$

Percent reduction in variance: $100(1 - 0.12) = \mathbf{88}\%$.

Interpretation: By accepting only the 0.1% most balanced randomizations (out of the chi-squared distribution), we reduce the variance of covariate imbalances by 88%. The distribution of $\overline{X}_T - \overline{X}_C$ under rerandomization (solid curve) is dramatically narrower than under pure randomization (dashed curve), centred tightly around zero for every covariate.

What v_a tells you. v_a is a property of the *design*, not the data. It tells you how much tighter the covariate imbalance distribution is under rerandomization compared to pure randomization. If covariates are strongly correlated with Y, this translates into a similarly large reduction in the variance of $\hat{\tau}$ itself.

6 Why Rerandomization Improves Precision

Rerandomization does not change the formula for $\hat{\tau}$. What it changes is the **distribution of** W, by ruling out the highly imbalanced assignments. To see why this improves precision, consider the following decomposition.

Approximate the control potential outcome as a linear function of covariates:

$$Y_i(0) \approx \alpha + X_i^\top \beta + \varepsilon_i, \qquad Y_i(1) = Y_i(0) + \tau,$$

where ε_i is the part of $Y_i(0)$ not explained by X_i. Then:

$$\hat{\tau} = \tau \ + \underbrace{(\bar{X}_T - \bar{X}_C)^\top \beta}_{\text{imbalance term}} + \underbrace{(\bar{\varepsilon}_T - \bar{\varepsilon}_C)}_{\text{residual term}}.$$

Under pure randomization, both terms contribute to $\mathrm{Var}(\hat{\tau})$:

$$\mathrm{Var}(\hat{\tau}) \approx \mathrm{Var}\big[(\bar{X}_T - \bar{X}_C)^\top \beta\big] + \mathrm{Var}(\bar{\varepsilon}_T - \bar{\varepsilon}_C).$$

Rerandomization forces M to be small, which constrains $\bar{X}_T - \bar{X}_C$ to be close to zero (in the Mahalanobis sense). This directly reduces $\mathrm{Var}[(\bar{X}_T - \bar{X}_C)^\top \beta]$ without affecting the residual term.

When does rerandomization help most? When the covariates are strongly predictive of the outcome (i.e. β is large). In that case, covariate imbalance contributes substantially to $\mathrm{Var}(\hat{\tau})$, and eliminating extreme imbalances produces a large precision gain. If covariates have no relationship with the outcome ($\beta \approx 0$), the imbalance term is negligible and rerandomization offers little precision benefit.

Example: Suppose pre-test score is strongly correlated with the outcome (post-test score). A random assignment that puts all high pre-test scorers in the treated group will make $\hat{\tau}$ noisy: it is unclear whether any difference in outcomes is due to the treatment or to pre-existing ability. Rerandomization rules out exactly those assignments, producing a tighter sampling distribution for $\hat{\tau}$ centered around the truth.

7 Inference After Rerandomization

7.1 The Fisher Randomization Test (Adjusted)

To compute a p-value after rerandomization, you cannot use the full set of $\binom{N}{N_T}$ treatment assignments as the reference distribution. You must restrict to the *acceptable* ones:

1. Compute the observed test statistic T^{obs} (e.g. $\hat{\tau}^{obs} = \overline{Y}_T^{obs} - \overline{Y}_C^{obs}$).
2. Repeatedly draw treatment assignments $W^{(b)}$, $b = 1, \dots, B$.
3. **Keep only those $W^{(b)}$ that satisfy** $\varphi_M(X, W^{(b)}) = 1$ (i.e. $M^{(b)} \leq a$).
4. For each accepted $W^{(b)}$, compute the test statistic $T^{(b)}$ under the sharp null $H_0 : Y_i(1) = Y_i(0)$ for all i.
5. The p-value is the proportion of accepted $T^{(b)}$ that are at least as extreme as T^{obs}.

Why is this valid? Under the sharp null, all potential outcomes are fixed. The only randomness is in W. By restricting to acceptable $W^{(b)}$, we are computing the probability of observing T^{obs} or more extreme, given that we *would have* accepted the randomization. This exactly mirrors the design: we committed to only use acceptable randomizations, so the reference set for inference should be only acceptable randomizations.

7.2 Neymanian Inference

Neymanian (confidence interval based) inference is more complex after rerandomization. Li, Ding, and Rubin (2018) showed that the asymptotic distribution of $\hat{\tau}$ under rerandomization is *not* normal, even in large samples. Instead, it is a **mixture of truncated normal distributions**: rerandomization truncates the distribution of $\overline{X}_T - \overline{X}_C$, and if X is correlated with Y, this truncation propagates to $\hat{\tau}$.

Practical implication. You cannot use a standard t-interval after rerandomization. Confidence intervals require specialised methods that account for the truncated-normal asymptotic distribution. The Fisher randomization test (adjusted as above) remains the simplest valid approach for inference.

8 Worked Example: Checking a Randomization

Suppose we have $N = 8$ units, $N_T = N_C = 4$, and $K = 2$ covariates: age (X_1) and baseline score (X_2). The covariate matrix and one candidate randomization are:

Unit i	W_i	Age X_{i1}	Baseline X_{i2}
1	1	25	60
2	1	30	70
3	1	28	55
4	1	35	80
5	0	22	50
6	0	40	75
7	0	38	65
8	0	27	45

Step 1: Compute the covariate means by group.

$$\overline{X}_T = \begin{pmatrix} 29.5 \\ 66.25 \end{pmatrix}, \qquad \overline{X}_C = \begin{pmatrix} 31.75 \\ 58.75 \end{pmatrix}, \qquad \overline{X}_T - \overline{X}_C = \begin{pmatrix} -2.25 \\ 7.5 \end{pmatrix}.$$

Step 2: Compute $\mathrm{cov}(X)$ from all 8 units.

$$\mathrm{cov}(X) = \begin{pmatrix} 37.5 & 82.1 \\ 82.1 & 137.5 \end{pmatrix}, \qquad [\mathrm{cov}(X)]^{-1} \approx \begin{pmatrix} 0.134 & -0.080 \\ -0.080 & 0.037 \end{pmatrix}.$$

Step 3: Compute M.

$$\frac{1}{1/N_T + 1/N_C} = \frac{1}{1/4 + 1/4} = 2.$$

$$M = 2 \times (-2.25 \quad 7.5) \begin{pmatrix} 0.134 & -0.080 \\ -0.080 & 0.037 \end{pmatrix} \begin{pmatrix} -2.25 \\ 7.5 \end{pmatrix}.$$

Computing the inner product step by step:

$$\begin{pmatrix} 0.134 & -0.080 \\ -0.080 & 0.037 \end{pmatrix} \begin{pmatrix} -2.25 \\ 7.5 \end{pmatrix} = \begin{pmatrix} (0.134)(-2.25) + (-0.080)(7.5) \\ (-0.080)(-2.25) + (0.037)(7.5) \end{pmatrix} = \begin{pmatrix} -0.902 \\ 0.458 \end{pmatrix}.$$

$$(-2.25)(-0.902) + (7.5)(0.458) = 2.03 + 3.44 = 5.47.$$
$$M = 2 \times 5.47 = 10.94.$$

Step 4: Accept or reject?

If we set $p_a = 0.10$ with $K = 2$ covariates, then $a = F^{-1}_{\chi^2_2}(0.10) = 0.211$.

Since $M = 10.94 \gg 0.211$, this randomization is **rejected**. The groups are badly imbalanced on the joint distribution of age and baseline score. We return to Step 3 and draw a new randomization.

Sanity check: what does a small M look like? If the two groups had nearly identical means $\overline{X}_T \approx \overline{X}_C$, then $\overline{X}_T - \overline{X}_C \approx \mathbf{0}$, and $M \approx 0$. The closer M is to zero, the more balanced the randomization. We accept whenever $M \leq a$, i.e. whenever the imbalance falls in the most balanced $100p_a\%$ of all possible randomizations.

9 Summary: How Rerandomization Fits In

	Pure Randomization	Blocking	Rerandomization
Pre-experiment covariate use	None	Discrete groups	Continuous (M)
Balance guaranteed	On average	Exactly within blocks	Up to threshold a
Number of covariates	Any	Few (1–2 typically)	Many (via M)
Inference	Standard	Standard	Modified
Requires knowing which X matter	No	Yes	No

Conclusions.

- Rerandomization improves covariate balance between groups, giving the researcher more confidence that an observed treatment effect is genuinely causal and not a confound of pre-existing group differences.
- If the covariates being balanced on are correlated with the outcome, rerandomization also *increases precision* in estimating τ: the variance of $\hat{\tau}$ under rerandomization is at most v_a times the variance under pure randomization.
- The key cost is inferential complexity: standard t-tests and normal-approximation confidence intervals are no longer valid. The analysis must match the design.

10 Practice Problems

Problem 4. You are designing a small experiment ($N = 6$, $N_T = N_C = 3$) with $K = 1$ covariate (age in years):

Unit	Age
A	22
B	25
C	30
D	35
E	40
F	45

A candidate randomization assigns units A, C, E to treatment and B, D, F to control.

(a) Compute $\overline{X}_T - \overline{X}_C$ for this randomization.

(b) With $K = 1$, the Mahalanobis distance simplifies to $M = (\overline{X}_T - \overline{X}_C)^2/[(1/N_T + 1/N_C)\, s_X^2]$, where s_X^2 is the sample variance of all six ages. Compute s_X^2 and then M.

(c) With $p_a = 0.10$ and $K = 1$: $F^{-1}_{\chi^2_1}(0.10) = 0.016$. Should this randomization be accepted or rejected?

Solution

(a) Compute $\overline{X}_T - \overline{X}_C$.

Treatment group: units A, C, E with ages 22, 30, 40.

$$\overline{X}_T = \frac{22+30+40}{3} = \frac{92}{3} \approx 30.67.$$

Control group: units B, D, F with ages 25, 35, 45.

$$\overline{X}_C = \frac{25+35+45}{3} = \frac{105}{3} = 35.$$

$$\overline{X}_T - \overline{X}_C = 30.67 - 35 = -4.33.$$

The treated group is on average about 4.33 years younger than the control group.

(b) Compute s_X^2 and then M.

The sample variance uses all six ages: 22, 25, 30, 35, 40, 45. The overall mean is:

$$\bar{X} = \frac{22+25+30+35+40+45}{6} = \frac{197}{6} \approx 32.83.$$

The squared deviations are:

$$(22-32.83)^2 = 117.3, \quad (25-32.83)^2 = 61.3, \quad (30-32.83)^2 = 8.0,$$

$$(35-32.83)^2 = 4.7, \quad (40-32.83)^2 = 51.4, \quad (45-32.83)^2 = 148.1.$$

Summing and dividing by $N-1=5$:

$$s_X^2 = \frac{117.3+61.3+8.0+4.7+51.4+148.1}{5} = \frac{390.8}{5} = 78.17.$$

Now compute M using the $K=1$ formula:

$$\frac{1}{N_T} + \frac{1}{N_C} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

$$M = \frac{(\overline{X}_T - \overline{X}_C)^2}{\left(\frac{1}{N_T} + \frac{1}{N_C}\right) s_X^2} = \frac{(-4.33)^2}{\frac{2}{3} \times 78.17} = \frac{18.75}{52.11} \approx 0.360.$$

(c) Accept or reject?

The threshold is $a = F^{-1}_{\chi^2_1}(0.10) = 0.016$.

Since $M = 0.360 > 0.016$, this randomization is **rejected**.

The value $M = 0.360$ sits well above the threshold, meaning this level of age imbalance (-4.33 years) is more extreme than what we are willing to accept. Only the 10% most balanced randomizations pass ($M \leq 0.016$), and this is not one of them. We return to Step 3 and draw a new treatment assignment.

Note: the threshold $a = 0.016$ is very strict — it corresponds to the bottom 10% of the χ^2_1 distribution, which has most of its mass away from zero. In practice, for $K=1$, this means only randomizations with near-perfect age balance are accepted.